EMCH 574 – Vibration and Waves
Homework 03

Student J.C. Vaught
Due date TBD

Contents

Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items
- UG students are not required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement reference
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga...) reference
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the y-axis scaled between -1 and +1. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and Abbreviations

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

1 String vibration

A.1 General Analysis Of String Vibration

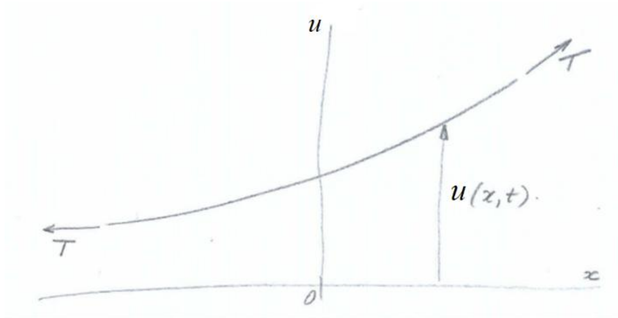


Figure 1 Taut string under tensile load T: (a) general view; (b) infinitesimal element

Consider a taut string of mass m per unit length stretched by a constant tension load T (Figure 1). The string undergoes small- amplitude transverse vibration with frequency ω as described by the following formula: $u(x,t) = U(x)e^{-i\omega t}$ (vibration motion) (1) Do the following:

(1) state the assumption made about the string tension T during the vibration motion u

(2) draw free-body diagram (FBD) of an infinitesimal string element of length ds and write sum of y forces in terms of string tension T and slope α

(3) apply small-displacement approximation and write the approximate expressions for ds, $\sin\alpha$, $\sin(\alpha + d\alpha)$

(4) write the linearized expression for the sum of y forces

(5) express the slope increment $d\alpha$ in terms of space increment dx and the relevant partial derivative of the displacement $u(x,t)$

(6) apply Newton's law of motion (NLM) and write the NLM equation

(7) write the governing equation for string vibration in terms of tension T, mass per unit length m, and the relevant partial derivatives of the displacement $u(x,t)$

(8) define wavespeed c for wave traveling in a taut string in terms of tension T and mass per unit length m

(9) write the wave equation

(10) substitute Eq. (1) into item (9) and eliminate the time dependency $e^{-i\omega t}$ to obtain an ODE defining $U(x)$

(11) recall the definition of the wavenumber γ and substitute into (10) to obtain an ODE in terms of only γ and $U(x)$

(12) assume exponential solution of the form $U(x) = e^{sx}$

(13) develop the characteristic equation for s

(14) solve the characteristic equation and write the roots s_1, s_2

(15) substitute (14) into (12) and write the general solution for $U(x)$ in terms of constants A_1, A_2

and complex exponentials depending on γ

(16) use Euler's formulae to rewrite (15) in terms of new constants C_1, C_2 and trigonometric functions depending on γ

(17) substitute (16) into Eq. (1) and write the general solution for $u(x,t)$ in terms of constats C_1, C_2, trigonometric functions of γx and a complex exponential function of ωt

(18) Graduate Work: write the general solution for $u(x,t)$ in terms of complex exponential functions of γ and ω

A.2 Analyze Natural Frequencies And Modeshapes Of A Vibrating String

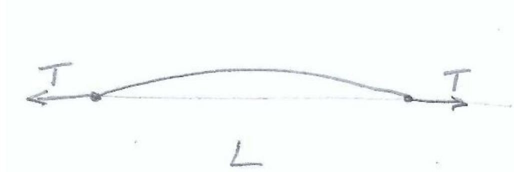


Figure 2 Taut string of length L under constant tension T

Consider a taut string of length L and mass m per unit length. The sting is pinned at both ends and stretched by a constant tension load T (Figure 2). The string undergoes transverse vibration with frequency ω. Do the following:

(1) write the boundary conditions at the two ends of string

(2) recall from Section A.1 the general solution for $u(x,t)$ expressed the multiplication of two terms:

- **(a)** first term: a function $U(x)$ depending on constants C_1, C_2 and trigonometric functions of γx
- **(b)** second term: a complex exponential function of ωt

(3) substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2

(4) solve (3) for C_2 and obtain an equation for C_1

(5) discuss the trivial solution

(6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(7) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z

(8) solve the eigenvalue equation and write the general expression for the eigenvalues z_j , $j = 1, 2, 3$,...

(9) use (8) to write the general expression for the wavenumber γ_j in terms of j and length L

(10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c

(11) solve (10) for frequency ω in terms of wavenumber γ and wavespeed c

(12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and string length L

(13) use (12) to write the general expression of the Hz frequency f_j in terms of j, wavespeed c, and string length L

(14) explain the meaning of fundamental frequency and overtones

(15) recall from HW01 the relationship between wavelength λ and wavenumber γ

(16) use (9) into (15) to write the general expression for the wavelength λ_j in terms of j and string length L

(17) use (16) to explain which wavelength multiples could accommodate vibratory standing waves in a string of length L fixed at both ends

(18) recall from (2)(a) the expression for $U(x)$

(19) use (4) to eliminate C_2

(20) set $C_1 = 1\text{in}$ (19) and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j

(21) give the expression of modeshape orthonormality

A.3 Calculate String Vibration

Consider a taught string of length $L = 22\,\mathrm{in}$ tuned to the note G3. The purpose is to calculate the string vibration characteristics and the modeshapes for $j = 1, \ldots, N$ modes with $N = 5$. We should also verify mode orthonormality using numerical integration on a $\mathrm{linspace}(0, L, N_x)$ with $N_x = 1000$. Recall the appropriate results from in-class instruction class and BB notes to do the following:

(1) Display input data

(2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string.

(3) Use frequency f_1 and length L to calculate the wavespeed c

(4) Calculate and display modal wavenumbers γ_j and frequencies ω_j

(5) Calculate the modeshapes, plot each mode separately and then plot overlapped all the N modes in a single plot

(6) Verify modeshape orthonormality by numerical integration

(7) Discuss your results

A.4 Calculate D'Addario Guitar Strings

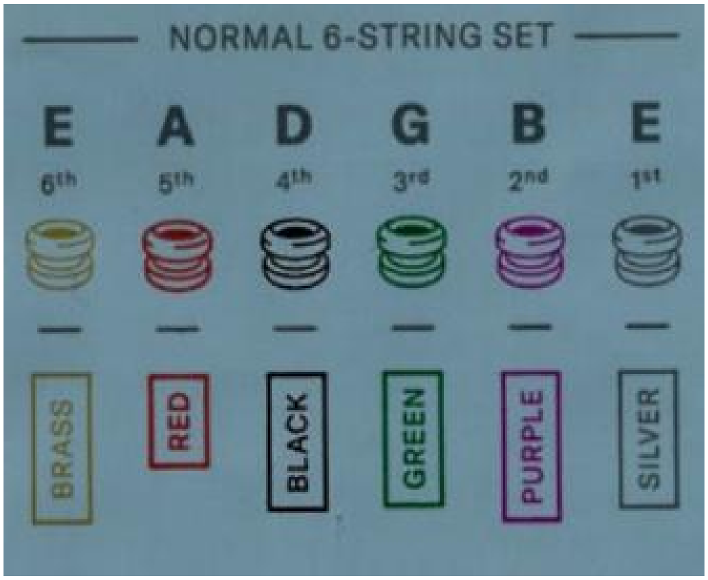


Table 1

Brass Red Black Green Purple Silver

musical

E **A D G B** E

note M, grams 6.42 4.18 2.45 1.332 0.655 0.367 T, lbf 24.95 28.93 29.93 30.05 23.31 23.36

Consider a set of six light-gauge d'Addario guitar strings. All six strings have the same length $L = 25.0\,\text{in}$. The mass M and the tension T of each string are given in in Table 1.

(1) Display input data For each of the six strings, do the following:

(2) Display string length L in m

(3) Calculate and display distributed mass, m, in kg/m

(4) Display tension T in N

(5) Calculate wavespeed c in m/s

(6) Calculate fundamental frequency f_1 in Hz

(7) Find the musical note that most closely approximates the frequency and give its letter name and octave number (e.g., G3 signifies note G in the 3rd octave)

(8) Display the frequency of the musical notes found at (7)

(9) Calculate the percentage difference between your findings and the nominal frequency of that musical note

(10) Comment on your findings and identify possible sources of error.

(11) Graduate work: Calculate the necessary adjustments in mass, tension, or both to achieve frequency match within 0.00%

A.5 Calculate Vibration Of A Plucked String

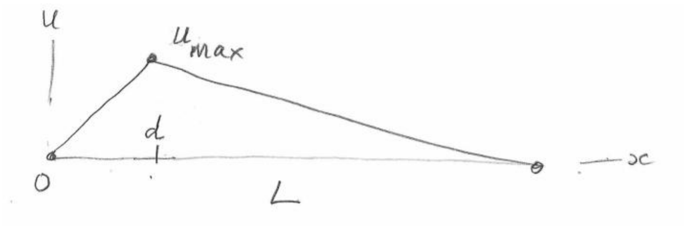


Figure 3 Taut string plucked at an arbitrary location d

Consider a taught string of length $L = 25.5\,\mathrm{in}$ tuned to the note G3. The string is plucked by applying an initial displacement $u_{\max} = 1.25\,\mathrm{mm}$ at a location d. In this problem, we will consider three possible plucking locations: $d = L/2$, $L/4$, $7L/10$. The purpose is to calculate the response $u(x,t)$ of the plucked string using the modal expansion method with $N = 10$. Recall the appropriate results from in-class instruction class and BB notes to do the following:

(1) Display input data

(2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding fundamental mode period τ_1.

(3) Use frequency f_1 and length L to calculate the wavespeed c

(4) Calculate modal wavenumbers γ_j and frequencies ω_j

(5) Calculate and plot overlapped the N modes

(6) For each plucking location, do the following:

- **(a)** display the value d and plot the initial shape of the string $u_0(x)$
- **(b)** calculate by numerical integration the modal participation factors A_j, display A_j in mm, and do a bar plot
- **(c)** plot the time response of the string at the plucking location $u(d,t)$ for $t = 0, \ldots, t_{\max}$ with $t_{\max} = 3\tau_1$
- **(d)** Graduate work: reconstruct the initial string shape and plot it overlapped on the actual shape

(7) Discuss comparatively the results for the three plucking locations

A.6 Calculate Frequency Response Of A Struck String

This problem addresses the frequency response of a taut string when struck by a point load like in the case of the hammered dulcimer (Figure 4).

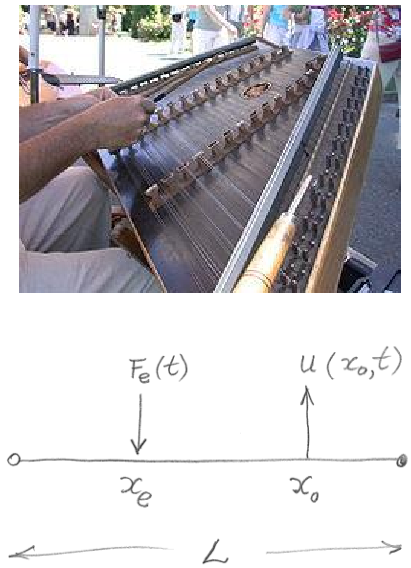


Figure 4 Struck string: (a) the hammered dulcimer; (b) schematic

Consider a taught string of length $L = 25.5\,\text{in}$ and distributed mass $m = 0.0015\,\text{kg/m}$ tuned to the note G3. The string damping ratio is assumed to have same value $\zeta_0 = 2.1\%$ in all modes. The string is struck at location x_e. The response of the string is measured with a piezoelectric acoustic pickup at location x_0. Three x_e, x_0 cases are considered: case x_e x_0

- **(a)** $L/3$ $L/3$
- **(b)** $L/3$ $L/2$
- **(c)** $L/2$ $L/2$ The striking force is assumed harmonic with amplitude $F_e = 0.5\,\text{N}$ and frequency range $f_{\text{start}} = 20\,\text{Hz}$ through $f_{\text{end}} = 2000\,\text{Hz}$ with $N_f = 5000$ steps. The objective is to calculate and plot the frequency response

function (FRF) over the frequency range using modal expansion method with $N = 10$ modes. Do the following:

(1) Display input data

(2) State the note name and octave and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding period τ_1 and the wavespeed c

(3) For $j = 1, \ldots, N$, calculate and display the modal properties wavenumber γ_j, frequency ω_j, mass m_j; damping ζ_j and plot overlapped the modeshapes

(4) Define the frequency range f using $\text{linspace}(f_{\text{start}}, f_{\text{end}}, N_f)$ and then define $\omega = 2\pi f$

(5) For each x_e, x_0 case, plot the $|\text{FRF}(\omega; x_e, x_0)|$ in dB and find the frequencies and nearest musical notes, as well as the index of the string modal frequency

(6) Graduate work: plot the time response of the normalized sound picked up at $x_0 = L/2$

when the string was struck at $x_e = L/3$. The excitation frequency should be the dominant note frequency in the FRF plot. The time plot should cover three periods of the dominant frequency.

2 Flexural vibration

B.1 General Analysis Of Flexural Vibration

Consider a uniform beam of mass m per unit length and bending stiffness EI undergoing flexural motion with vertical displacement $w(x,t)$ as shown in Figure 5. Centroidal axes are assumed.

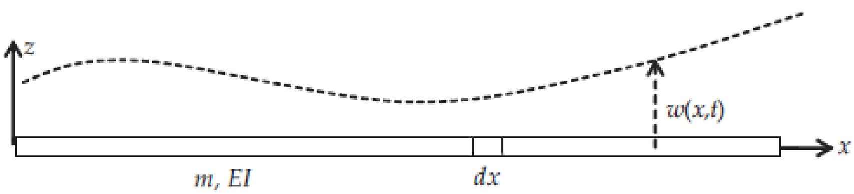


Figure 5 Beam undergoing flexural vibration

Do the following:

(1) state the assumptions needed for performing the flexural vibration analysis

(2) draw free-body diagram (FBD) of an infinitesimal beam element dx and show the stress resultants N, M, V.

(3) write the definitions of the stress resultants N and M

(4) use stress-strain relation to write the stress resultants in terms of strains

(5) perform kinematic analysis to express the stress resultant M in terms of partial derivatives of the displacement w

(6) perform free-body analysis of the infinitesimal element dx and find the sum of forces ΣF_z and sum of moments ΣM

(7) apply Newton's law of motion (NLM) in the z direction and obtain an equation for the shear force V

(8) apply NLM in the rotational direction and obtain an equation for the moment M

(9) substitute (6) into (8) and get an equation that relates M and V

(10) use (5) into (9) and obtain the governing equation

(11) define the flexural constant a in terms of E, I, m

(12) use (11) into (10) to obtain the governing equation as a PDE in x, t, and flexural constant a

(13) assume harmonic vibration of the form $w(x,t) = W(x)e^{-i\omega t}$

(14) substitute (13) into (12) and obtain an ODE in x with constants a, ω

(15) define the wavenumber γ in terms of ω and a and substitute in (14) to write the ODE with only one constant, γ

(16) assume $W(x)$ as an exponential function of x of the form $= e^{sx}$

(17) substitute (16) into (15) and derive the characteristic equation

(18) find the four roots of the characteristic equation

(19) write the general solution $W(x)$ in terms of constants A_1, A_2, A_3, A_4 and complex exponential functions of γx

(20) rewrite the general solution (19)(13) in terms of constats C_1, C_2, C_3, C_4 multiplying trigonometric and hyperbolic functions of γx

(21) Graduate Work: explain how one can find the constants C_1,

C_2, C_3, C_4

B.2 Analyze Cantilever Beam Vibration

A cantilever beam is fixed at one end and free at the other end. Consider a cantilever beam of length L as shown in Figure 6. The beam has mass density per unit length m, elastic modulus E, and cross-section second moment of area I.

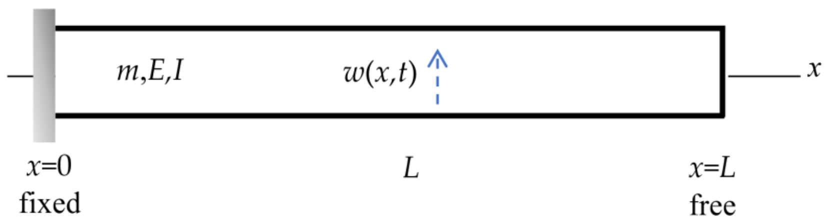


Figure 6 Vibration of a cantilever beam of length L. The cantilever beam is fixed at one end and free at the other end

Do the following:

(1) write the boundary conditions at the two ends of the bar

(2) recall from the notes the general solution $w(x, t)$ consisting of the multiplication of two terms:

- **(a)** first term: a function $W(x)$ depending on constants C_1, C_2, C_3, C_4, and trigonometric and hyperbolic functions of γx
- **(b)** second term: a complex exponential function of ωt

(3) substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2, C_3, C_4

(4) solve (3) for C_3, C_4 in terms of C_1, C_2 and obtain a set of equations for C_1, C_2

(5) discuss the trivial solution

(6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(7) introduce the notation $z = \gamma L$ and write the transcendental eigenvalue equation in terms of z

(8) use the MATLAB function fzero to solve the transcendental eigenvalue equation and find z_j, $j = 1, \ldots, N$ with $N = 5$

(9) use item (7) to write the general expression for the wavenumber γ_j in terms of z_j and length L

(10) Recall from Problem B.1 the definition of wavenumber γ in terms of frequency ω and flexural constant a

(11) solve (10) for frequency ω in terms of wavenumber γ and flexural constant a

(12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of eigenvalue z_j, beam constant a, and beam length L

(13) recall definition of a in terms of bending stiffness EI and mass per unit length m and obtain an expression for the natural frequencies ω_j and f_j in terms of z_j, EI, m, L

(14) use (4) to obtain an expression for the ratio C_1/C_2 in terms of wavenumber γ and beam length L. Denote this expression to be $-\beta$

(15) recall from (2)(a) the expression for $W(x)$

(16) use (4) and (14) to write the constants C_1, C_3, C_4 in terms of C_2

(17) substitute (16) into (15) and write $W(x)$ in terms of C_2, γx, β

(18) Set $C_2 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j and β_j

(19) give the expression of modeshape orthonormality

(20) Graduate work:

(a) write the approximate expression for the eigenvalues z_j with $j \geq 6$

(b) solve the transcendental equation at item (7) for $N = 10$ and compare with the prediction using the approximate expression of item (a)

B.3 Calculate Cantilever Beam Vibration

Assume a cantilever beam of length $L = 445\,\mathrm{mm}$ and cross-section dimensions $h = 3\,\mathrm{mm}$, $b = 37\,\mathrm{mm}$. The material is steel with $E = 200\,\mathrm{GPa}$ and $\rho = 7850\,\mathrm{kg/m^3}$. Do the following:

(1) display input data

(2) calculate the cross-section area A

(3) calculate mass per unit length m

(4) calculate second moment of area I

(5) calculate flexural stiffness EI

(6) calculate flexural constant a

(7) solve the transcendental equation defined in Problem B.2 item (7) to find z_j, $j = 1, \ldots, N$ with $N = 5$

(8) calculate the wavenumbers γ_j

(9) calculate angular frequencies ω_j

(10) calculate Hz frequencies f_j

(11) calculate the modeshape coefficients β_j

(12) calculate the modeshapes $W_j(x)$, plot each mode separately and then plot overlapped all the N modes on a single plot

(13) verify modeshape orthonormality by numerical integration

(14) comment on your results

B.4 Calculate The 512 C Tuning Fork

Consider the 512 C aluminum-alloy tuning fork (Figure 7). Common values for aluminum elastic modulus and density are $E = 70\,\mathrm{GPa}$ and $\rho = 2700\,\mathrm{kg/m^3}$.This tuning fork is stated to give the middle C note (C_5) according to the scientific pitch scale reference In the scientific pitch scale, the middle C note C_5 has the frequency $f_1 = 512\,\mathrm{Hz}$.

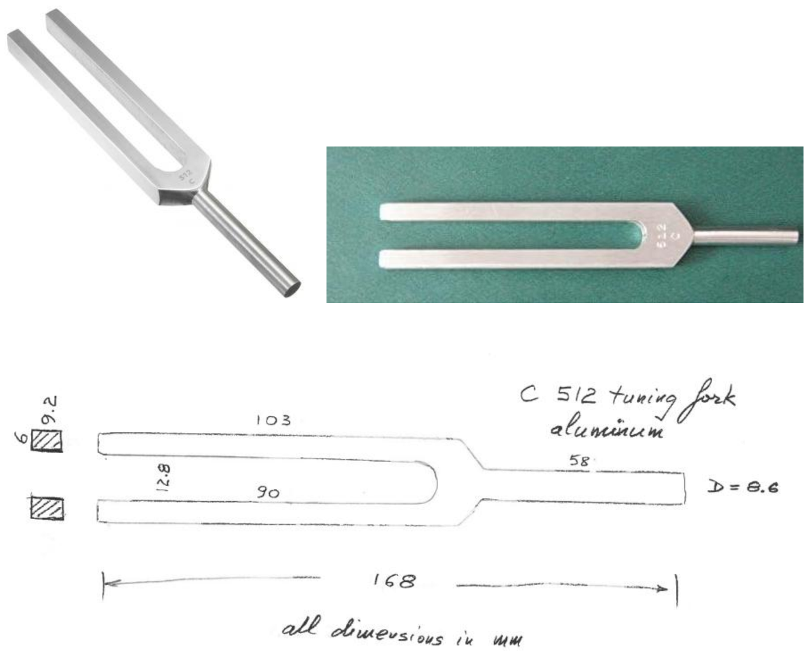


Figure 7 512 C tuning fork reference Tuning-Fork/dp/B00BLBE8QE . The sketch is done by DrG by measuring the actual fork.

Use the cantilever-beam analysis developed in Problem B.2 to perform the engineering evaluation of this tuning fork. Do the following:

(1) display input data

(2) use the sketch in Figure 7 to estimate the cantilever-beam length L. Hint: use engineering intuition to estimate where the root of the cantilever beam could be assumed

(3) use the sketch in Figure 7 to obtain the cantilever-beam cross- section dimensions h, b

(4) calculate cross-section area A

(5) calculate mass per unit length m

(6) calculate second moment of area I

(7) calculate flexural stiffness EI

(8) calculate flexural constant a

(9) use the results of Problem B.3 to calculate the fundamental frequency f_1 in Hz

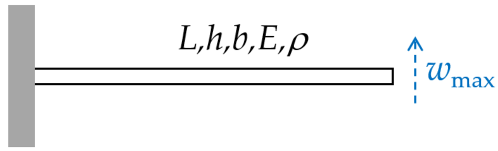
(10) compare the calculated f_1 with the nominal tuning fork frequency of $512\,\mathrm{Hz}$, calculate the percentage difference, and comment on your results

(11) manually adjust the estimated beam length to a new value L_{new} until the new frequency $f1_{\text{new}}$ equals $512\,\text{Hz}$

(12) comment on the engineering estimation and adjustment process developed in the solution of this problem

(13) Graduate work: find the frequencies of the next three higher harmonics of the tuning fork after length adjustment

B.5 Calculate Vibration Of A Plucked Reed



Consider a cantilever beam of length $L = 14\,\mathrm{mm}$ and cross-section dimensions $h = 0.3\,\mathrm{mm}$, $b = 0.75\,\mathrm{mm}$. The material is steel with $E = 200\,\mathrm{GPa}$ and $\rho = 7850\,\mathrm{kg/m^3}$. The beam is plucked with an initial tip displacement $w_{\mathrm{max}} = 0.15\,\mathrm{mm}$. The purpose is to calculate the response $w(x,t)$ of the plucked beam using the modal expansion method. Recall the appropriate results from class and/or BB notes and do the following:

(1) Display input data

(2) Calculate cross-section area A, mass per unit length m, cross- section second moment of area I, flexural stiffness EI, and flexural constant a

(3) Calculate the roots of the cantilever eigenvalue equation for $j = 1,\dots,N$ with $N = 5$

(4) Calculate the fundamental Hz frequency f_1 and state the symbol (letter and octave) of the nearest musical note. Calculate the fundamental mode period τ_1

(5) Recall the formulas for cantilever beam modeshapes and plot the modes $j = 1,\dots,N$

(6) calculate and plot the initial deformed shape of the beam $w_0(x)$

(7) Assume the modal expansion method, calculate the modal coefficients A_j, $j = 1,\dots,N$, and do a bar plot

(8) plot the time response of the tip of the beam $w(L,t)$ for three fundamental-mode periods

(9) Comment on your results

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Flexural Vibration In A Free-Free Beam

(1) sketch a free-free beam indicating all the information needed for the analysis

(2) write the boundary conditions at the two ends of the beam in terms of displacement $w(x,t)$ and stress resultants $M(x,t)$, $V(x,t)$

(3) repeat all the steps of Section B.2

C.2 Calculate Wind-Chime Tones

Consider the wind chime shown in Figure 8 and Figure 9.

Figure 8 Wind chime

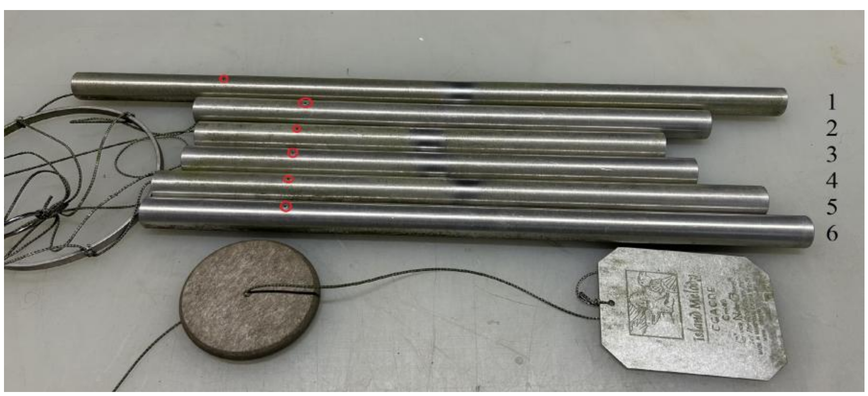


Figure 9 Tubes of the wind chime with the suspension points circled in red.

The wind chime tubes are made of aluminum with $E = 70\,\text{GPa}$ and $\rho = 2700\,\text{kg/m}^3$. The outer and inner diameters of the tubes are $D = 22.3\,\text{mm}$; $d = 19.4\,\text{mm}$. The tube length is L. Each tube is suspended at a distance e from its end. The L and e values are given in Table 2.

Table 2: The measured L and e values of the six wind chime tubes # L, mm e, mm 1 500 110 2 352 78 3 312 70 4 332 74 5 384 87 6 406 90

Use the free-free beam analysis developed in Problem C.1 to perform the engineering evaluation of the wind chime tubes. Do the following:

(1) calculate the cross-section area A

(2) calculate mass per unit length m

(3) calculate second moment of area I

(4) calculate flexural stiffness EI

(5) calculate flexural constant a

(6) use the MATLAB function fzero to find the first root of the transcendental eigenvalue equation defined in Problem C.1

(7) use the length values given in Table 2 to calculate the wavenumber γ and the corresponding frequencies ω_1 and f_1 for each of the six tubes

(8) use reference to estimate the musical notes that might be associated with the frequencies of the six tubes

(9) Graduate work:

- **(a)** calculate and plot the modeshape $W_1(x)$ for each of the six tubes
- **(b)** estimate the location x_0 where the modeshape $W_1(x)$ crosses the zero line, compare the values x_0 with the values e given in Table 2, and calculate percentage difference. Comment on your results

(10) Graduate work: find by manual iterations the adjusted tube length values that will give exact musical notes without # at item (8) above

C.3 Calculate Flexural Vibration In A Free-Free Beam

Assume a free-free beam of length $L = 445\,\mathrm{mm}$, and cross-section dimensions $h = 3\,\mathrm{mm}$, $b = 37\,\mathrm{mm}$. The material is steel with $E = 200\,\mathrm{GPa}$ and $\rho = 7850\,\mathrm{kg/m^3}$. Do the following:

Repeat the steps of Section B.3

C.4 Analyze Flexural Vibration Of Simply Supported

BEAM

(1) sketch a simply supported beam indicating all the information needed for the analysis

(2) write the boundary conditions at the two ends of the beam in terms of displacement $w(x,t)$ and stress resultants $M(x,t)$, $V(x,t)$

(3) recall from class notes the general solution $w(x,t)$ consisting of the multiplication of two terms:

- **(a)** first term: a function $W(x)$ depending on constants C_1, C_2, C_3, C_4, and trigonometric and hyperbolic functions of γx
- **(b)** second term: a complex exponential function of ωt

(4) substitute (3) into (2) and write boundary conditions in terms of constants C_1, C_2, C_3, C_4

(5) solve (4) for C_2, C_4 and obtain a set of equations for C_1, C_3

(6) discuss the trivial solution

(7) choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(8) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z

(9) write the general expression of the roots z_j of the eigenvalue equation

(10) use item (8) to write the general expression for the wavenumber γ_j in terms of j and length L

(11) Recall from class notes the definition of wavenumber γ in terms of frequency ω and flexural constant a

(12) solve (11) for frequency ω in terms of wavenumber γ and flexural constant a

(13) use (10) and (12) to write the general expression of the natural frequency ω_j in terms of j, beam constant a, and beam length L

(14) recall definition of a in terms of bending stiffness EI and mass per unit length m and obtain an expression for the natural frequencies ω_j and f_j in terms of j, EI, m, L

(15) use item (5) to solve for C_3

(16) substitute the values found so far for C_2, C_3, C_4 and write $W(x)$ in terms of C_1 and γx

(17) Set $C_1 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j

(18) give the expression of modeshape orthonormality

C.5 Calculate Simply Supported Flexural Vibration

Assume a simply supported beam of length $L = 445\,\mathrm{mm}$, and cross- section dimensions $h = 3\,\mathrm{mm}$, $b = 37\,\mathrm{mm}$. The material is steel with $E = 200\,\mathrm{GPa}$ and $\rho = 7850\,\mathrm{kg/m^3}$. Do the following: Repeat the steps of Section B.3

C.6 Analysis Of Wave Equation In A Slender Bar

Assume a uniform slender bar of cross section A and density ρ (Figure 10). The bar undergoes time-varying axial displacement $u(x,t)$. The displacement is assumed uniform across the bar cross section.

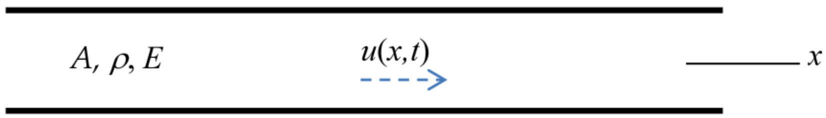


Figure 10 Wave motion in a uniform slender bar

Do the following:

(1) state the assumption made about motion and deformation in the slender bar

(2) draw free-body diagram (FBD) of an infinitesimal bar element of length dx and write Newton's law of motion (NLM) equation

(3) write elasticity equation that relates stress and displacement

(4) substitute (3) into (2) and derive the partial differential equation (PDE) in terms of displacement

(5) define the wavespeed c in terms of elasticity modulus E and density ρ

(6) write the wave equation

(7) assume separation of x and t variables and write the solution as a product between an x function $U(x)$ and an exponential function in t of the form $e^{-i\omega t}$

(8) substitute (7) into (6) and derive an ordinary differential equation (ODE) in terms of c, ω, $U(x)$

(9) recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c and use it to simplify (8)

(10) assume $U(x)$ as an exponential function of x of the form e^{sx}

(11) substitute (10) into (9) and derive the characteristic equation

(12) find the roots s_1, s_2 of the characteristic equation in terms of the wavenumber γ and write the general solution $U(x)$ in terms of complex exponentials of γx

(13) add the time dependency defined at (7) and write the general solution for $u(x,t)$ in terms of complex exponentials of γx, ωt, and constants A_1, A_2

(14) rewrite the general solution (13) in terms of constants C_1, C_2, trigonometric functions of γx and a complex exponential function of ωt

(15) Graduate Work: identify the forward and backward waves in the general solution derived at (13)

C.7 Calculate Axial Vibration With Various Boundary Conditions

Consider a slender aluminum bar of diameter $D = 20\,\mathrm{mm}$ and length $L = 3.0\,\mathrm{m}$. Common values for aluminum elastic modulus and density are $E = 70\,\mathrm{GPa}$ and $\rho = 2700\,\mathrm{kg/m^3}$. The bar undergoes axial vibration with frequency ω. The purpose is to calculate the string vibration characteristics, the modeshapes, for $j = 1, \ldots, N$ modes with $N = 5$. We should also verify mode orthonormality using numerical integration on a $\mathrm{linspace}(0, L, N_x)$ with $N_x = 1000$. Three boundary condition cases are considered:

boundary conditions case $x = 0$ $x = L$

- **(a)** fixed fixed
- **(b)** fixed free
- **(c)** free free

(1) Display input data

(2) Calculate wavespeed c Then, for each of the three cases, do the following:

(3) Display boundary conditions

(4) Calculate and display modal vibration characteristics

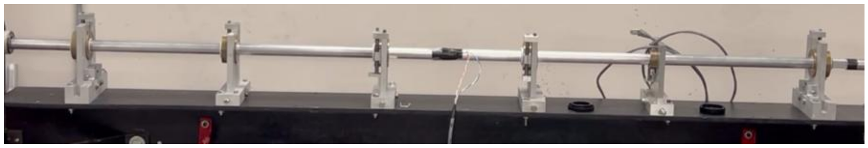
(5) Calculate and plot overlapped the displacement modeshapes

(6) Verify modeshape orthonormality by numerical integration

(7) Calculate and plot overlapped the normalized stress modeshapes

(8) Discuss your results

C.8 Calculate The Hopkinson Bar



reference

The Hopkinson pressure bar is used to create and measure axial stress pulses. It consists of a long and slender metallic bar which is hit at one end. The hit creates a stress pulse that propagates in the bar and is measured by strain gauges. A common application is to measure the compression behavior of a small specimen placed in what is called the Split-Hopkinson bar. Consider an aluminum Hopkinson pressure bar of diameter $D = 20\,\text{mm}$. Common values for aluminum elastic modulus and density are $E = 70\,\text{GPa}$ and $\rho = 2700\,\text{kg/m}^3$. Do the following:

(1) Calculate the axial wavespeed c in aluminum

(2) Assume the bar length is $L = 3.2\,\text{m}$ and calculate the time t required to travel the length of the bar

(3) Calculate the natural frequencies f_1, f_2, f_3 of the first three modes of bar axial vibration

(4) Calculate and plot the corresponding axial vibration modeshapes

(5) Graduate Work: Research the topic of the split Hopkinson pressure bar and write a short description of this apparatus and its uses

C.9 Analyze Axial Vibration In A Free-Free Bar

Consider a slender bar of finite length L as shown in Figure 11. The bar has cross-section area A, density ρ, elastic modulus E. The bar is free at both ends.

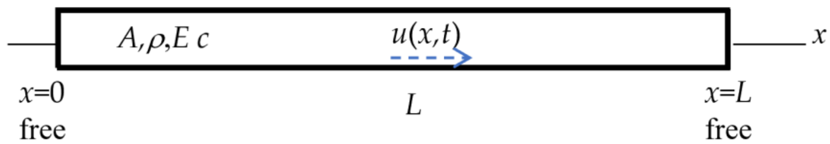


Figure 11 Vibration of a slender bar of finite length L free at both ends

Do the following:

(1) write the boundary conditions at the two ends of the bar

(2) recall from Problem C.6 the general solution $u(x, t)$ in terms of:

- **(a)** first term: a function $U(x)$ depending on constants C_1, C_2 and trigonometric functions of γx
- **(b)** second term: a complex exponential function of ωt

(3) substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2

(4) solve (3) for C_1 and obtain an equation for C_2

(5) discuss the trivial solution

(6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution

(7) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z

(8) solve the eigenvalue equation and write the general expression for the eigenvalues z_j , $j = 1, 2, 3$,...

(9) use (8) to write the general expression for the wavenumber γ_j in terms of j and length L

(10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c

(11) use (10) to solve for frequency ω in terms of wavenumber γ and wavespeed c

(12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and bar length L

(13) use (12) to write the general expression of the Hz frequency f_j in terms of j, wavespeed c, and length L

(14) explain the meaning of fundamental frequency and overtones

(15) recall from HW01 the relationship between wavelength λ and wavenumber γ

(16) use (9) into (15) to write the general expression for the wavelength λ_j in terms of j and length L

(17) use (16) to explain which wavelength multiples could be accommodated vibratory standing waves in a bar of length L fixed at both ends

(18) recall $U(x)$ from (2)(a) and use (4) to eliminate C_1

(19) set $C_2 = 1$ and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j

(20) give the expression of modeshape orthonormality

(21) write the general expression of the stress modeshapes

C.10 Analyze Axial Vibration In A Fixed-Fixed Bar

Do the following:

(1) sketch a bar fixed at both ends indicating all the information needed for the analysis

(2) write the boundary conditions at the two ends of the bar

(3) repeat the steps of Problem C.9

C.11 Analyze Axial Vibration In A Fixed-Free Bar

Do the following:

(1) sketch a bar fixed at one end and free at the other end indicating all the information needed for the analysis

(2) write the boundary conditions at the two ends of the bar

(3) repeat the steps of Problem C.9